

Sara Klaasen

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Education

PhD	ETH Zurich , Geophysics - Eidgenössische Technische Hochschule Zürich (ETH Zürich, Swiss Federal Institute of Technology Zurich) - ETH medal	Zurich, Switzerland 2020 – 2025
Msc	ETH Zurich , Geophysics	Zurich, Switzerland 2018 – 2020
Bsc	University College Roosevelt (UCR) , Liberal Arts and Sciences - Magna cum Laude - Major in Earth sciences, biology, mathematics and physics	Middelburg, Netherlands 2014 – 2017

Experience

Australian National University , Postdoctoral researcher Researching DAS projects in New Zealand. Imaging Taupo Volcano.	Canberra, Australia 2026 – present
ETH Zurich , Postdoctoral researcher	Zurich, Switzerland 2025 – 2026
Sport, Recreatie en Onderwijsvoorzieningen (SRO) , Admin Online administration	Haarlem, Netherlands 2017 – 2017
Provinciale Zeeuwse Courant (PZC) , Online editor - Online content management of the provincial newspaper - Writing and editing news articles	Vlissingen, Netherlands 2015 – 2017

Awards

ETH Medal ETH Zurich The ETH Medal is given to outstanding doctoral theses (2k CHF).	2026
Outstanding Student Presentation Award (OSPA) Seismological Society of America Photonics Meeting	2024

Grants

Swiss National Science Foundation (SNF) postdoc.mobility grant Swiss National Science Foundation Grant to fund a 2-year postdoc position (136k CHF)	2025
Women-in-STEM grant Fix the Leaky Pipeline program, ETH Zürich Grant for peer-mentoring group (2k CHF).	2021
VSBFonds grant VSBFonds Grant to support a Masters programme abroad (10k EUR).	2018

Research collaborations

Australian Research Council (ARC) Centre of Excellence in Optical Microcombs for Break-through Science (COMBS) Associate investigator www.combs.org.au	2026 – present
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Seismicity on Venus: Prediction and Detection

2022 – 2024

Early Career Researcher (ECR) member

teams.issibern.ch/seismicityonvenus

International Space Science Institute (ISSI) international team.

Teaching experience

Data Analysis and Visualisation with Python in Earth Sciences

2022 – 2025

Teaching assistant

Bachelor course at ETH Zürich

Geophysical field lab

2021 – 2023

Teaching assistant

Bachelor fieldcourse at ETH Zürich

Dynamische Erde I

2020 – 2023

Guest lecturer

Annual ten minute lectures about current geophysical applications during the Bachelor course at ETH Zürich.

Research Output

BOOK CHAPTERS

The leading edge of volcanology: Advances and innovations in monitoring and interpreting volcanic activity

2027

Michael P Poland, Elske de Zeeuw-van Dalftsen, Koki Aizawa, **Sara Klaasen**, Fiona D'Arcy, Emma J Nicholson, Kye Anderson

Elsevier

Book chapter submitted to "The encyclopedia of Volcanoes, Third Edition". Chapter has been accepted.

Fiber optic sensing for volcano monitoring and imaging volcanic processes

2025

Philippe Jousset, Gilda Currenti, Shane Murphy, Eva Eibl, Corentin Caudron, Lise Retailleau, Christopher Wollin, **Sara Klaasen**, Takeshi Nishimura, Andreas Fichtner, Zack Spica, Arnaud Lemarchand, Charlotte Krawczyk

Springer

doi.org/10.1007/978-3-031-86841-2_17

Book chapter in Modern Volcano Monitoring

JOURNAL ARTICLES

Submarine seismicity monitoring with distributed acoustic sensing near Santorini and Kolumbo Volcano

2026

Sara Klaasen, Thomas Hudson, Sanchit Sachdeva, Paraskevi Nomikou, Andreas Fichtner

Seismica

doi.org/10.26443/seismica.v5i1.1845

Seismic constraints on glacier density

2025

Ariane Lanteri, Scott Keating, Lars Gebraad, **Sara Klaasen**, Marta Pienkowska-Cote, Olaf Eisen, Andrea Zunino, Kristín Jónsdóttir, Coen Hofstede, Dimitri Zigone, Andreas Fichtner

Scientific Reports

doi.org/10.1038/s41598-025-26440-z

The Seismogenic Thickness of Venus

2025

Julia Maia, Ana-Catalina Plesa, Iris van Zelst, Richard Ghail, Anna Gülcher, Mark Panning, Sven Peter Näsholm, Barbara de Toffoli, Anna Horleston, Krystyna Smolinski, **Sara Klaasen**, Robert Herrick, Raphael Garcia

Journal of Geophysical Research: Planets

doi.org/10.1038/s41598-025-26440-z

- Towards a widely applicable earthquake detection algorithm for fibreoptic and hybrid fibreoptic-seismometer networks 2025
Thomas Hudson, **Sara Klaasen**, Olivier Fontaine, Conor Bacon, Kristin Jonsdottir, Andreas Fichtner
Geophysical Journal International
doi.org/10.1093/gji/ggae459
- An illustrated guide to: Distributed and integrated fibre-optic sensing in seismology 2024
Andreas Fichtner, Fabian Walter, Patrick Paitz, **Sara Klaasen**, Daniel Bowden, Sebastian Noe, Nils Müller, Dominik Husmann, Jacques Morel
Earthquake Sciences
dx.doi.org/10.1016/j.eqs.2024.09.006
- Seismic wave detectability on Venus using ground deformation sensors, infrasound sensors on balloons and airglow imagers 2024
Raphael Garcia, Iris van Zelst, Taichi Kawamura, Sven Peter Näsholm, Anna Horleston, **Sara Klaasen**, Maxence Lefevre, Celine Marie Solberg, Krystyna Smolinski, Ana-Catalina Plesa, Quentin Brissaud, Julia Maia, Simon Stähler, Philippe Lognonné, Mark Panning, Anna Gülcher, Richard Ghail, Barbara de Toffoli
Earth and Space Science
doi.org/10.1029/2024EA003670
- Modelling uncertainty in P-wave arrival-times retrieved from DAS data: case-studies from 15 fibre optic cables 2024
Emanuele Bozzi, Nicola Piana Agostinetti, Andreas Fichtner, **Sara Klaasen**, Arantza Ugalde, Biondo Biondi, S Yuan, Torsten Dahm, Marius Isken, Patrick Paitz, Fabian Walter, Alain Baird, Carlos Becerril, Takeshi Nishimura, Junzhu Shen, Tieyuan Zhu, Gilberto Saccorotti
Geophysical Journal International
doi.org/10.1093/gji/ggae364
- Complex spatial distribution of onset amplitude and waveform correlation: case studies from different DAS experiments 2024
Emanuele Bozzi, Gilberto Saccorotti, Nicola Piana Agostinetti, Carlos Becerril, Andreas Fichtner, **Sara Klaasen**, Takeshi Nishimura, Junzhu Shen, Fabian Walter, Tieyuan Zhu
Bulletin of geophysics and oceanography
doi.org/10.4430/bgo00458
- Challenges in submarine fiber-optic earthquake monitoring 2024
Jonas Igel, **Sara Klaasen**, Sebastian Noe, Paraskevi Nomikou, Konstantinos Korantzalos, Andreas Fichtner
Journal of Geophysical Research: Solid Earth
doi.org/10.1029/2024JB029556
- Subglacial volcano monitoring with fibre-optic sensing: Grímsvötn, Iceland 2023
Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristín Jónsdóttir, Lars Gebraad, Patrick Paitz, Andreas Fichtner
Volcanica
doi.org/10.30909/vol.06.02.301311
- Fiber-optic observation of volcanic tremor through floating ice sheet resonance 2022
Andreas Fichtner, **Sara Klaasen**, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Patrick Paitz, Kristin Jonsdottir
The Seismic Record
doi.org/10.1785/0320220010
- Distributed acoustic sensing in volcano-glacial environments — Mount Meager, British Columbia 2021
Sara Klaasen, Patrick Paitz, Nadja Lindner, Jan Dettmer, Andreas Fichtner
Journal of Geophysical Research: Solid Earth
doi.org/10.1029/2021JB022358

Detecting Seismic Events with Computer Vision: Applications for Fiber-Optic Sensing Solvi Thrastarson, Robert Torfason, Sara Klaasen, Patrick Paitz, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner ESS Open Archive doi.org/10.1002/essoar.10509693.1	2021
FIRST-AUTHOR CONFERENCE PROCEEDINGS	
Modelling months of deformation during the Sundhnúkur eruptions on the Reykjanes Peninsula, Iceland Sara Klaasen, Daniel Bowden, Kristin Jonsdottir, Yesim Cubuk-Sabuncu, Jack Maxwell, Meghan Miller, Andreas Fichtner Seismological Society of America (SSA) optical seismology and the next era of seismic sensing	2026
Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner International Association of Volcanology and Chemistry of the Earth's Interior (IAVCEI) General Meeting	2025
Full-Waveform Inversion of DAS data on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner IAVCEI General Meeting	2025
Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner SSA General Meeting	2025
Moment Tensor inversion with Full-Waveform Inversion and Distributed Acoustic Sensing on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner European Geosciences Union (EGU) Copernicus Meetings	2024
Submarine volcano monitoring with fiber-optic cables at Santorini Sara Klaasen, Paraskevi Nomikou, Konstantinos Karantzalos, Jonas Igel, Andreas Fichtner American Geophysical Union (AGU) Annual Meeting	2023
Going towards Full Waveform Inversion (FWI) with Distributed Acoustic Sensing (DAS) on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Jonas Igel, Andreas Fichtner AGU Annual Meeting	2023
Volcano monitoring and modelling using Distributed Acoustic Sensing (DAS) on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Sebastian Noe, Andreas Fichtner International Union of Geodesy and Geophysics (IUGG) General Assembly	2023
Monitoring glacially covered volcanoes with Fibre-Optic Sensing Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner IAVCEI General Meeting	2023
Volcanic Event Characterization with Fiber-Optic Sensing Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner AGU Annual Meeting	2022
Fibre-Optic Sensing for Volcano Monitoring on Grímsvötn, Iceland Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner EGU General Assembly	2022

Distributed Acoustic Sensing on Volcanoes: Grímsvötn and Fagradalsfjall, Iceland Sara Klaasen , Solvi Thrastarson, Kristin Jonsdottir, Yesim Cubuk-Sabuncu, Patrick Paitz, Andreas Fichtner AGU Annual Meeting	2021
Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia Sara Klaasen , Patrick Paitz, Jan Dettmer, Andreas Fichtner SSA General Meeting	2021
Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia Sara Klaasen , Patrick Paitz, Jan Dettmer, Andreas Fichtner EGU General Assembly	2021
Distributed Acoustic Sensing for the Exploration of the Mount Meager Volcanic Complex, British Columbia, Canada Sara Klaasen , Andreas Fichtner, Patrick Paitz, Jan Dettmer AGU Annual Meeting	2020

Invited Talks

Keck observatory workshop, Pre-conference workshop • SSA optical seismology and the next era of seismic sensing	Hawaii, United States of America (USA) 2026
Frontiers in Fiber-Optics Sensing in Seismology, Conference session • International Association of Seismology and Physics of the Earth's Interior-International Association of Geomagnetism and Aeronomy (IASPEI-IGA)	Lisbon, Portugal 2025
ECR Keynote talk, Conference keynote • IAVCEI	Geneva, Switzerland 2025
Seminar, Seminar • Stanford University	San Francisco, USA 2025
Seminar, Seminar • University of Washington	online 2024
An innovative photonic vision of volcanoes and geothermal systems, Conference Session • SSA Photonic Seismology	Vancouver, Canada 2024
Colloquium, Seminar • Commonwealth Scientific and Industrial Research Organisation (CSIRO)	Perth, Australia 2023
Geophysics Weekly Meeting, Seminar • Australian National University	Canberra, Australia 2023
Seminar, Seminar • University of Canterbury	Christchurch, New Zealand 2023
Workshop on Distributed Acoustic Sensing, Pre-conference workshop • AGU	online 2021

Service

Convening conference sessions, IAVCEI, AGU

Peer-review, Seismica, Seismological Research Letters (SRL)

Languages

Dutch, native

English, C2

German, B2

French, B1